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REVIEW **Open Access**

Advancing Climate-Resilient Sorghum: the Synergistic Role of Plant Biotechnology and Microbial Interactions

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Abstract

Climate-related problems such as drought stress, extreme temperature, erratic rainfall patterns, soil degradation, heat-waves, flooding, water logging, pests and diseases afflict the production and sustainability of sorghum. These challenges may be addressed by adopting climate-resilient practices and using advanced agronomic techniques. These challenges are being addressed through innovative applications of plant biotechnology and microbiology, which offer targeted solutions to enhance sorghum's resilience. For instance, biotechnological tools like CRISPR/Cas9 enable precise genetic modifications to improve drought and heat tolerance, while microbial inoculants, such as plant growth-promoting rhizobacteria (PGPR) and arbuscular mycorrhizal fungi (AMF), enhance nutrient uptake and stress tolerance through symbiotic interactions. However, biotechnological tools lead to the development of sorghum varieties with heat, drought and salinity tolerance, while marker-assisted selection significantly accelerates breeding for stress-resilient traits. When genetic engineering is introduced, genes encoding heat shock proteins, Osmo protectants and antioxidant pathways are introduced to increase plant resistance to abiotic stress. These compounds stabilise cellular structures, protect enzymes, and maintain osmotic balance, enhancing the plant's ability to survive and function in adverse environmental conditions. At the same time, it is reported that microbiology offers beneficial microbes, nitrogen-fixing bacteria, phosphate-solubilizing microorganisms, and arbuscular mycorrhizal fungi that help enhance nutrient availability, soil health and water uptake. Combinations of endophytes and microbial inoculants enhance plant immunity to pests and diseases while increasing tolerance to stress. Biocontrol agents such as *Bacillus* and *Trichoderma* contain suppression of pathogens and need less dependence on the use of chemical pesticides. On top of that, genetic modification increases the nutritional quality of sorghum biofortified. This is where biotechnology and microbiology work together to deliver sustainable farming systems reducing environmental impacts, boosting yields and securing food supply under environmental stresses. This review aims to examine the synergistic integration of plant biotechnology and microbial interactions as a strategy to enhance sorghum's resilience to climate-induced stresses, including drought, elevated temperatures, and nutrient-deficient soils. It highlights recent advancements in biotechnological tools such as gene editing, marker-assisted selection, and tissue culture, alongside the emerging

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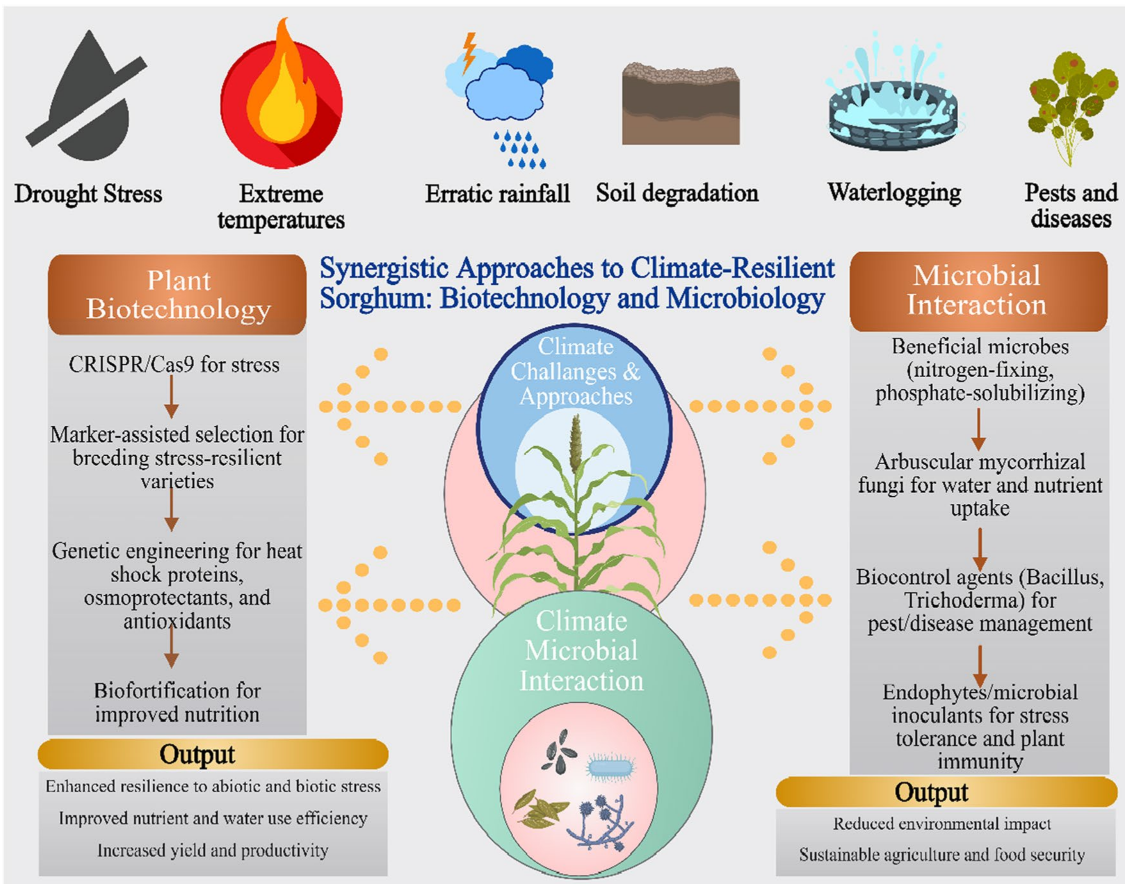
role of plant-beneficial microbes in promoting stress tolerance and improving soil health. By synthesizing current knowledge across these disciplines, this review seeks to outline a framework for future research that harnesses the intersection of biotechnology and microbial ecology to support the sustainable improvement of sorghum resilience.

Keywords Sorghum, Climate resistant, Plant biotechnology, Microbial interactions

Graphical Abstract



Introduction

Sorghum (*Sorghum bicolour* L. Moench) is a vital cereal crop renowned for its resilience to harsh environmental conditions, making it a cornerstone of food security in arid and semi-arid regions. As the fifth most cultivated cereal globally, sorghum plays a critical role in addressing the agricultural challenges posed by climate change, including drought, heat stress, and soil degradation (Teferra and Awika 2019). Originating from Africa, its cultivation has expanded to Asia, the Americas, and Australia, providing critical resources for human consumption, livestock feed, and biofuel production (Ananda et al. 2020). Still, the growing impacts of climate change, including unpredictable rainfall, warmer temperatures, and higher biotic stresses, pose significant challenges to the productivity and sustainability of sorghum (Table 1). To mitigate these challenges, innovative approaches that integrate plant biotechnology and microbial interactions have emerged as promising solutions. This review investigates the combined use of advanced biotechnological techniques, including CRISPR/Cas9, marker-assisted selection, and microbial inoculants, to improve sorghum's ability to withstand climate-induced challenges. By analyzing the relationship between genetic engineering and supportive microbial communities, we seek to offer a detailed

Table 1 Major Climate Stresses Impacting on Sorghum Growth

Climate Stress	Impact on Sorghum Growth	Mechanisms of Stress Impact	References
Drought	Drought significantly reduces sorghum productivity by limiting water availability, affecting grain yield and biomass	Reduced photosynthetic efficiency, dehydration, osmotic stress, reduced nutrient uptake	Seleiman et al. 2021; Abreha et al. 2022
Heat Stress	High temperatures, especially during flowering and grain filling, can lead to sterility, reduced grain size, and poor seed set	Disruption of cellular processes, reduced enzyme activity, and impaired pollen development	Sher et al. 2024; Sita et al. 2017
Salinity	Soil salinity impairs root function and nutrient uptake, leading to stunted growth, chlorosis, and reduced yield	Salt accumulation in plant tissues disrupts water balance and ion homeostasis, causing cellular toxicity	Muhammad et al. 2024; Atta et al. 2023
Flooding/Waterlogging	Excessive water reduces soil aeration, leading to oxygen deprivation in roots, which hinders growth and reduces yield potential	Oxygen deficiency in roots, root damage, and nutrient imbalance due to poor soil aeration	Li et al. 2019
Nutrient Deficiency	Sorghum growth is affected by low availability of essential nutrients, particularly nitrogen, phosphorus, and potassium, leading to stunted growth and poor grain development	Impaired photosynthesis, reduced metabolic activity, and stunted root development	Gladman et al. 2022
Pests and Pathogens	Pests (e.g., aphids, caterpillars) and pathogens (e.g., leaf blight, Fusarium) can damage plants, leading to yield loss and quality deterioration	Mechanical damage to plant tissues, reduced photosynthesis, and induction of plant defense responses	Al-Khayri et al. 2023
UV Radiation	Elevated UV radiation can damage plant cells, particularly affecting photosynthetic processes and leading to reduced growth and yield	DNA damage, oxidative stress, and inhibition of key metabolic pathways	Sachdev et al. 2021

guide for creating climate-resilient sorghum varieties, thereby promoting sustainable food production in a warming climate. Climate resilience in sorghum is driven by a combination of morphological, physiological, and molecular adaptations. These include an extensive root system, efficient stomatal regulation, and the accumulation of osmoprotectants, which are small organic molecules, such as proline, glycine betaine, and sugars, that accumulate in plant cells under stress conditions like drought, salinity, or extreme temperatures.

Advances in genetic engineering and molecular breeding have facilitated the development of sorghum cultivars with enhanced drought tolerance, pest resistance, and nutritional value (Xin et al. 2021). Biotechnological tools, such as CRISPR/Cas9 and transcriptomic analysis, have accelerated trait improvement and enabled precise enhancements of sorghum's performance under abiotic stress conditions. (Nigam et al. 2025). Simultaneously, microbial associations especially with endophytes and rhizosphere bacteria offer promising avenues for enhancing sorghum's resilience. Plant growth-promoting rhizobacteria (PGPR) and mycorrhizal fungi can improve nutrient uptake, induce systemic tolerance to drought and salinity, and enhance plant immunity (Aruna et al. 2024). Harnessing beneficial microbial interactions supports sustainable agriculture by reducing chemical use and enhancing crop productivity and soil health.

The synergistic integration of plant biotechnology with microbial biotechnology presents a transformative strategy for advancing climate-resilient sorghum. This holistic approach, supported by omics technologies and precision agriculture, holds promise for overcoming production constraints and meeting global food demands in a warming world. Collaborative research, policy support, and investment in biotechnology-driven agriculture are crucial for unlocking the full potential of sorghum in sustainable food systems. Sorghum is a staple crop in arid and semi-arid regions and is renowned for its drought tolerance and adaptability. However, the accelerating impacts of climate change pose significant challenges to its sustainable production. Rising global temperatures, erratic rainfall patterns, and increased occurrences of extreme weather events are projected to exacerbate abiotic and biotic stress factors, adversely affecting sorghum yields and quality (Njinju et al. 2022). Climate change significantly reduces soil moisture due to prolonged droughts. While sorghum is more drought-tolerant than other cereals, severe droughts can exceed its adaptive limits, resulting in diminished grain filling, poor panicle development, and reduced biomass. Additionally, heat stress during critical reproductive stages, especially flowering and grain-setting, can induce spikelet sterility, severely impacting yield potential (Table 2).

Climate change also intensifies soil salinity, affecting sorghum cultivation in regions where irrigation with saline water is prevalent. Salinity disrupts ion homeostasis and osmotic balance, impairing plant growth and metabolic functions (Liaqat et al. 2024). Additionally, elevated atmospheric carbon dioxide levels may alter sorghum's photosynthetic efficiency. Although increased CO_2 can enhance photosynthesis in C_4 crops like sorghum, the net benefits may be offset by increased water stress and nutrient limitations (Leakey et al. 2019). Biotic stresses, including the proliferation of pests and diseases, are expected to worsen under climate change scenarios. Higher temperatures and altered precipitation can expand the geographical range and lifecycle duration of sorghum pests such as stem borers and aphids, leading to more frequent and severe infestations (Okosun et al. 2021). Fungal pathogens causing grain mould and anthracnose are also likely to become more pervasive, further threatening yield stability and grain quality. Addressing these multifaceted impacts requires an integrative approach, combining advances in plant biotechnology and microbial interactions. Genetic engineering and molecular breeding techniques are being employed to develop sorghum varieties with enhanced drought, heat, and salt tolerance (Khalifa and Eltahir 2023). Complementarily, microbial inoculants, including plant growth-promoting rhizobacteria and endophytic fungi, can enhance sorghum's stress resilience by improving nutrient uptake, water-use efficiency, and stress hormone modulation (Chaffai et al. 2024). Harnessing these synergistic strategies offers a sustainable pathway for mitigating the adverse impacts of climate change on sorghum production and securing food systems in vulnerable regions.

Plant–microbe interactions are vital for sustainable agriculture, especially under abiotic and biotic stress. Beneficial microbial communities, including bacteria and fungi, enhance plant growth, improve nutrient acquisition, and induce tolerance to environmental challenges. In the context of climate-resilient crops like sorghum, leveraging these interactions offers promising avenues for sustainable yield enhancement and resource-efficient farming. Key players include plant growth-promoting rhizobacteria (PGPR), mycorrhizal fungi, and endophytic microbes which improve drought tolerance, nutrient solubilization, and pathogen resistance. PGPR strains, including *Azospirillum*, *Pseudomonas*, and *Bacillus* species, enhance nitrogen fixation, solubilize phosphorus, and produce phytohormones such as indole-3-acetic acid (IAA), thereby promoting plant growth and resilience under various stress conditions. Other well-studied PGPR species include *Rhizobium*, *Burkholderia*, *Enterobacter*, and *Serratia*,

Table 2 Overview of sorghum, emphasizing its importance, production, and nutritional significance

Aspect	Details	Region	Applications	Benefits	References
Importance	Climate-resilient cereal crop	Sub-Saharan Africa, India, USA	Food, fodder, fuel	Adaptable to drought, high temperatures, and low-fertility soils	Ouda et al. 2022
	Staple food source for millions	Global	Subsistence and commercial farming	Enhances food security in semi-arid regions	Mwamahonje et al. 2024
Production	5th most important cereal crop globally	India, Nigeria, Sudan, USA	Rainfed agriculture	Requires minimal input; high adaptability	Srinivasa et al. 2014
	Approx. 60 million tons produced annually	Worldwide	Food, feed, bioethanol	Integral to dryland farming systems	Tonapi et al. 2020
Macronutrient Value	High in carbohydrates (~ 72%), moderate protein (~ 11%), low in fat (~ 3%)	All cultivars	Human diet, animal feed	Provides energy; suitable for gluten-free diets	Eggleston et al. 2022
Micronutrient Value	Rich in iron, magnesium, phosphorus, B-complex vitamins	Varies by genotype	Nutritional supplementation	Helps reduce micronutrient malnutrition	Rather et al. 2023
Dietary Fiber	Good source of dietary fiber	Whole grain usage	Functional food industry	Improves gut health, lowers cholesterol, regulates blood sugar	Khalid et al. 2022
Bioactive Compounds	Contains polyphenols, tannins, and flavonoids	Pigmented varieties	Antioxidant supplements, health foods	Antioxidant, anti-inflammatory, anti-diabetic potential	Espitia-Hernández et al. 2022
Health Benefits	Helps manage diabetes, obesity, and celiac disease	Clinical nutrition	Therapeutic diets	Low glycemic index, gluten-free, rich in resistant starch	Khalid et al. 2022
Processing Uses	Used in flour, malt, popped grain, fermented foods	Food industry	Bakery, snack, beverage production	Offers texture and taste versatility; used in traditional and modern processed foods	Deribe and Kassa 2020
Climate Resilience	Survives in low rainfall and high-heat conditions	Arid and semi-arid regions	Dryland farming systems	Ensures yield stability under climate change	Rao et al. 2009
Gluten-Free Nature	Naturally gluten-free	Celiac-safe zones	Alternative to wheat	Safe for gluten-intolerant individuals	Hegde et al. 2023
Sustainability	Requires less water and fertilizer than wheat or maize	Global	Sustainable agriculture	Lower environmental footprint; soil-restoring crop	Hossain et al. 2022
Breeding Potential	High genetic diversity	Global germplasm collections	Crop improvement programs	Enables development of biofortified and climate-resilient varieties	Chaturvedi et al. 2022

which are also known for their abilities to fix atmospheric nitrogen, secrete organic acids for phosphate solubilization, and synthesize plant growth regulators including gibberellins, cytokinins, and IAA (Glick 2012; Teja et al. 2023; Chieb and Gachomo 2023). These microbes play a crucial role in sustainable agriculture by improving nutrient availability, stimulating root development, and enhancing plant tolerance to abiotic and biotic stresses. Mycorrhizal fungi, particularly arbuscular mycorrhizal fungi (AMF), enhance water uptake and facilitate nutrient exchange, significantly benefiting sorghum cultivation in water-limited environments (Birhanu and Negussie 2024). Biotechnology tools have revolutionized the understanding and manipulation of plant–microbe interactions. Advanced omics technologies, including metagenomics, metabolomics, transcriptomics, and proteomics, provide comprehensive insights into the microbial communities associated with sorghum and their functional roles in stress mitigation (Kimotho and Maina 2024). Genome editing tools such as CRISPR/Cas9 have been employed to improve sorghum's resilience by targeting specific genes related to abiotic stress tolerance (Rai et al. 2023). Moreover, synthetic biology approaches are being explored to engineer microbial consortia with enhanced synergistic effects, enabling customized solutions for varying agro-climatic conditions (Samantaray et al. 2024). Microbial inoculants, biofertilizers, and biostimulants derived from beneficial microorganisms represent scalable technologies for integrating plant–microbe interactions into mainstream agriculture. However, optimizing their efficacy requires understanding the complex dynamics between microbial strains, host genotypes, and environmental factors. Research on identifying region-specific and stress-adapted microbial strains holds significant potential for climate-resilient sorghum cultivation. This review provides a combination of plant biotechnology and microbial interactions to offer a dual strategy for enhancing sorghum's productivity and sustainability. Future research should prioritize integrative approaches that combine genetic improvements with microbial solutions to meet the growing demands for food security amid climate change.

Climate Challenges in Sorghum Cultivation

Abiotic Stresses

Abiotic stresses, particularly drought, heat, and salinity, represent major constraints on sorghum productivity, exacerbated by the changing climate. Sorghum is recognized for its inherent drought tolerance, yet prolonged and recurrent drought episodes significantly reduce biomass accumulation, grain yield, and seed viability. Recent studies have identified critical drought-responsive regulatory networks involving dehydration-responsive element-binding proteins (DREB), late embryogenesis abundant (LEA) proteins, and abscisic acid (ABA)-mediated signalling pathways (Kim et al. 2024). Marker-assisted selection and genomic selection have been increasingly employed to integrate these adaptive traits into new cultivars, improving water-use efficiency and osmotic adjustment capacities (Hualpa-Ramirez et al. 2024). Heat stress, particularly during the reproductive phase, leads to poor pollen viability, reduced kernel set, and lower grain fill. Studies by Khan (2020) highlight the role of heat-shock proteins (HSPs) and membrane lipid stability in maintaining cellular integrity under extreme temperatures. Moreover, sorghum's ability to sustain photosynthesis through modified leaf angle orientation and reduced transpiration rates is critical for developing heat-tolerant varieties (Ranawana et al. 2023). Integrating advanced biotechnological tools like gene pyramiding and transgenic methods offers promise for enhancing salinity tolerance in sorghum. Identifying sodium and potassium transporters (*HKT1*) and vacuole sequestration genes (*NHX1*) forms a genetic basis for breeding. Multidisciplinary strategies, including improved soil management and irrigation, are essential for implementation (Cao et al. 2022).

Biotic Stresses

Sorghum cultivation is increasingly challenged by biotic stresses, driven by the proliferation of pests and pathogens whose lifecycles are accelerated by rising global temperatures. Stem borers, shoot flies (*Atherigona soccata*), and aphids are among the most damaging pests. The invasive *Melanaphis sacchari* aphid, for example, causes significant losses by reducing photosynthetic efficiency and acting as a vector for viral diseases (Vasquez et al. 2024). Traditional pest control methods, including chemical pesticides, pose environmental risks, necessitating sustainable alternatives. Biological control using entomopathogenic fungi and parasitoids, coupled with genetic resistance, has shown promise (Zhou et al. 2024). Among diseases, anthracnose and grain mold complexes significantly reduce yield and grain quality. Anthracnose, caused by *Colletotrichum sublineola*, thrives in humid conditions and has shown increased prevalence under climate variability. Resistant sorghum varieties with major resistance genes (*ANTH1*) and quantitative trait loci (QTLs) for disease resistance are under development using marker-assisted breeding and CRISPR/Cas9 genome editing (Mengistu et al. 2021). In grain mold, fungal pathogens such as *Fusarium* and *Aspergillus* produce mycotoxins that threaten food safety. Integrating pre-harvest and post-harvest management practices with genetic improvements is essential for mitigating these risks (Wenndt 2020).

Socio-Economic Implications of Climate-Resilient Sorghum

The socio-economic impact of cultivating climate-resilient sorghum extends beyond mitigating food insecurity. In drought-prone regions of Sub-Saharan Africa, India, and other semi-arid regions, sorghum serves as a vital staple food and a primary source of fodder and income for millions of smallholder farmers (Tsumbu 2024). Enhancing sorghum's resilience to climate stresses can reduce household vulnerability and contribute to poverty reduction. Improved sorghum varieties with high grain quality, pest resistance, and adaptability to local conditions are increasingly recognized as tools for social transformation (Srinivasa 2024). Sorghum also has growing economic potential in industrial applications. Its use in bioethanol production and as a gluten-free grain for health-conscious consumers is expanding market opportunities. However, socio-economic benefits can only be realized through equitable access to improved seeds, farmer training on best agricultural practices, and supportive policy frameworks. Addressing barriers such as limited investment in research, weak extension services, and insufficient market linkages will be pivotal in maximizing the potential of climate-resilient sorghum. The future of sorghum cultivation depends on integrative approaches that blend conventional breeding, molecular biology, and agroecological strategies. Success scaling of these innovations requires robust collaborations among researchers, policymakers, and farming communities. Climate-smart practices, enhanced seed systems, and investment in biotechnology will collectively shape sustainable food systems in a warming world.

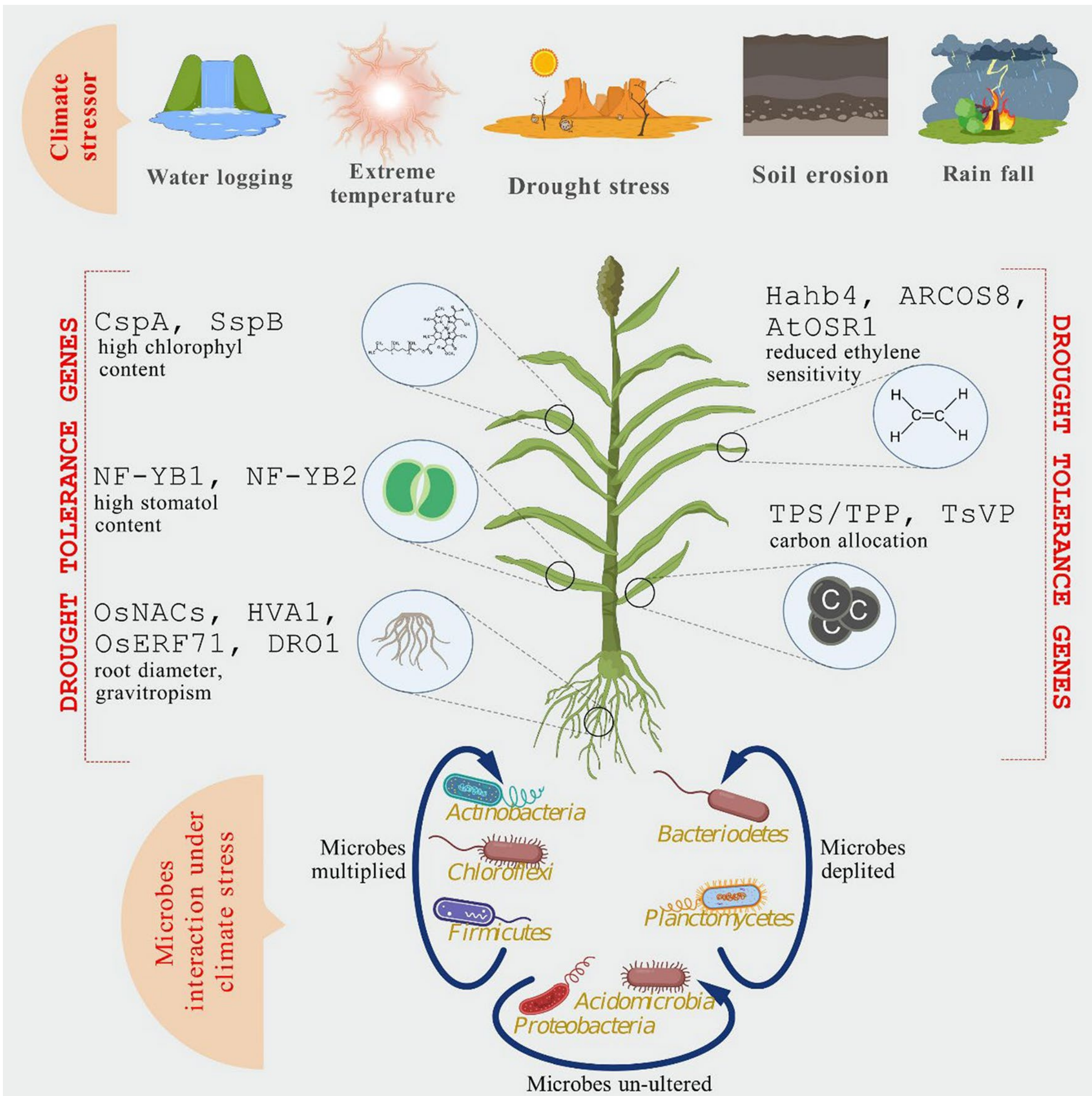


Fig. 1 The figure illustrates the impact of various climate stressors, including waterlogging, extreme temperature, drought stress, soil erosion, and rainfall, on plant physiology and associated microbial interactions. Drought tolerance genes play a crucial role in enhancing plant adaptation to stress conditions. Specific genes (CspA & SspB contribute to high chlorophyll content; NF-YB1 & NF-YB2 regulate stomatal density; OsNACs, HVA1, OsERF71, & DRO1 influence root diameter and gravitropism, supporting water acquisition under drought stress; Hahb4, ARCOS8, & AtOSR1 reduce ethylene sensitivity; TPS/TPP & TsVP contribute to carbon allocation for stress resilience. The lower section highlights the microbial response under climate stress. Microbial communities exhibit varied responses, with *Actinobacteria*, *Chloroflexi*, and *Firmicutes* multiplying, whereas *Bacteroidetes* and *Planctomycetes* experience depletion. Certain microbial groups, including *Acidomicrobia* and *Proteobacteria*, remain unaltered. This differential microbial interaction under stress conditions suggests a complex interplay between plant genetic responses and soil microbiota in climate adaptation

Role of Microbial Interactions in Sorghum Resilience

Sorghum is renowned for its drought and heat tolerance. The Fig. 1 depicts how climate stressors like waterlogging, extreme temperatures, drought, soil erosion, and rainfall affect plant physiology and microbial interactions. However, its productivity can be boosted by beneficial microorganisms. These microbial associations enhance nutrient availability, improved stress tolerance, and promote soil health. Recent studies emphasize the importance of plant-growth-promoting microorganisms (PGPMs), mycorrhizal fungi, nitrogen-fixing bacteria, and endophytes in boosting sorghum resilience, providing sustainable alternatives to chemical inputs (Fig. 2).

Plant-Growth-Promoting Microorganisms

PGPMs encompass a diverse group of bacteria and fungi that enhance plant growth by various mechanisms, including nutrient solubilization, phytohormone production, and abiotic stress mitigation. In sorghum, several bacterial species have shown promising effects on growth and drought resistance. For instance, *Azospirillum brasilense* promotes nitrogen fixation and root development, enabling sorghum to access deeper soil moisture during drought conditions (Kathait 2023). Similarly, *Bacillus subtilis* improves phosphorus uptake by producing organic acids that solubilize phosphate compounds, making them available to plants. It secretes various organic acids such as gluconic acid, citric acid, and oxalic acid that solubilize insoluble phosphate compounds, thereby increasing P bioavailability in the rhizosphere (Glick 2012). It also helps plants cope with stress by modulating hormones like auxins and abscisic acid (Nosheen et al. 2021). Besides *B. subtilis*, other PGPR such as *Pseudomonas*, *Azospirillum*, *Burkholderia*, and endophytes like *Penicillium* and arbuscular mycorrhizal fungi also improve phosphorus availability and stress tolerance through various mechanisms, including hormone

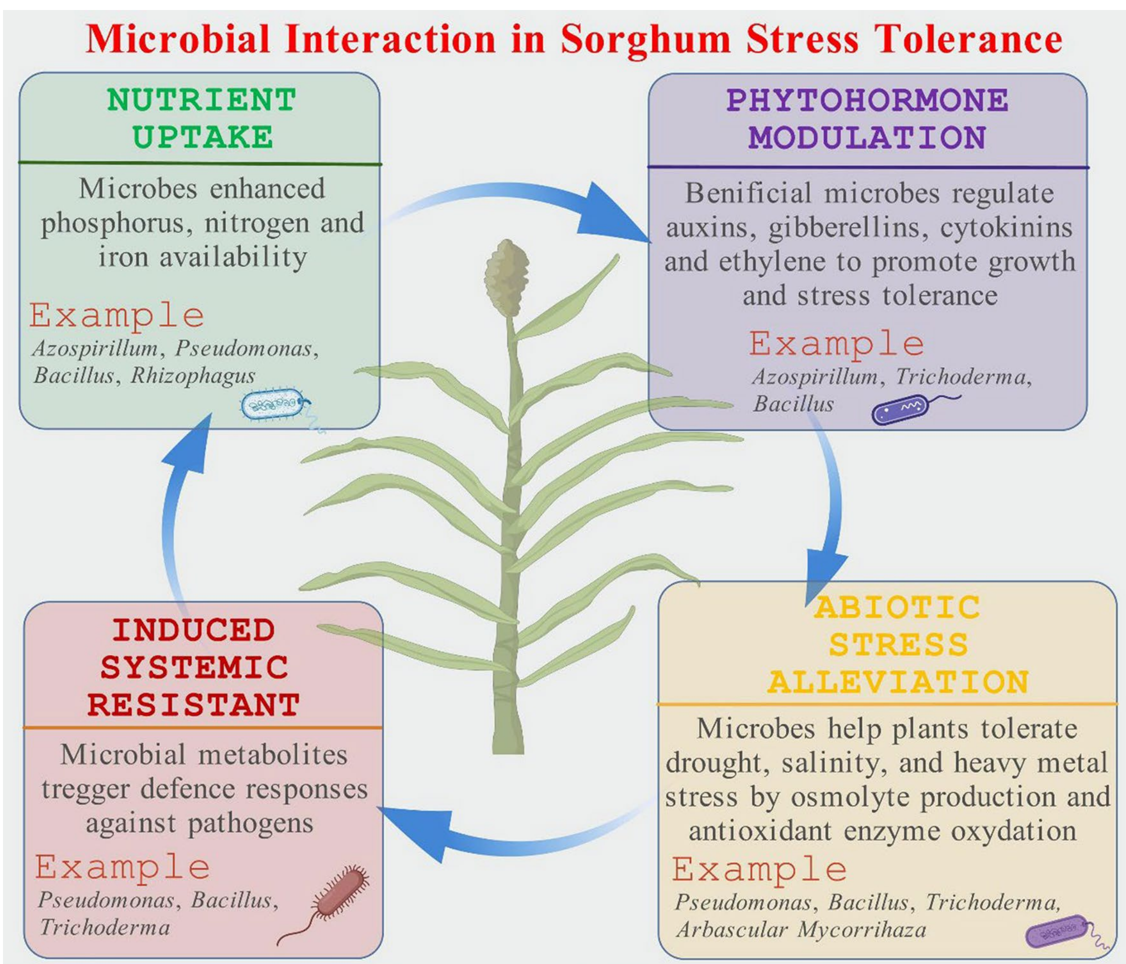


Fig. 2 This figure illustrates the types of microbial interaction in sorghum stress tolerance including nutrient uptake, phytohormonal modulation, induced systemic resistant and abiotic stress alleviation

regulation, antioxidant production, and root development support (Basu et al. 2021). Recent studies indicate that PGPMs also modulate the plant's hormonal balance under stress. Bacteria like *Pseudomonas fluorescens* secrete auxins and cytokinins, which promote root elongation and lateral root formation, crucial for water uptake (Karnwal and Kaushik 2011). Additionally, these bacteria produce siderophores, molecules that chelate iron from the soil and enhance its bioavailability to plants, addressing iron deficiencies that often occur in dry, alkaline soils (Ajinde et al. 2024). Inoculating sorghum with a combination of PGPMs has been shown to produce synergistic effects, resulting in higher biomass accumulation and improved grain yield even under nutrient-deficient and water-scarce environments (Teiba et al. 2023). This integrated microbial approach reduces dependence on chemical fertilizers and enhances sustainable agriculture practices.

Phosphate-solubilizing microorganisms (PSM) enhance phosphorus (P) availability and promote plant growth in sorghum through various biochemical mechanisms. They produce organic acids, such as citric and gluconic acids, which lower soil pH and dissolve insoluble phosphate compounds, releasing bioavailable P. PSM also secrete phosphatases and phytases, enzymes that mineralize organic phosphorus sources like phytate into plant-accessible forms. In addition to phosphorus solubilization, these microbes contribute to plant growth by synthesizing phytohormones (e.g., indole-3-acetic acid, jasmonic acid, and salicylic acid), fixing atmospheric nitrogen, and reducing ethylene levels through ACC deaminase activity, thereby mitigating stress responses. Furthermore, PSM enhances plant resilience by producing siderophores, lytic enzymes, and hydrogen cyanide (HCN), suppressing phytopathogens and improving overall plant health (Fig. 3). These combined mechanisms

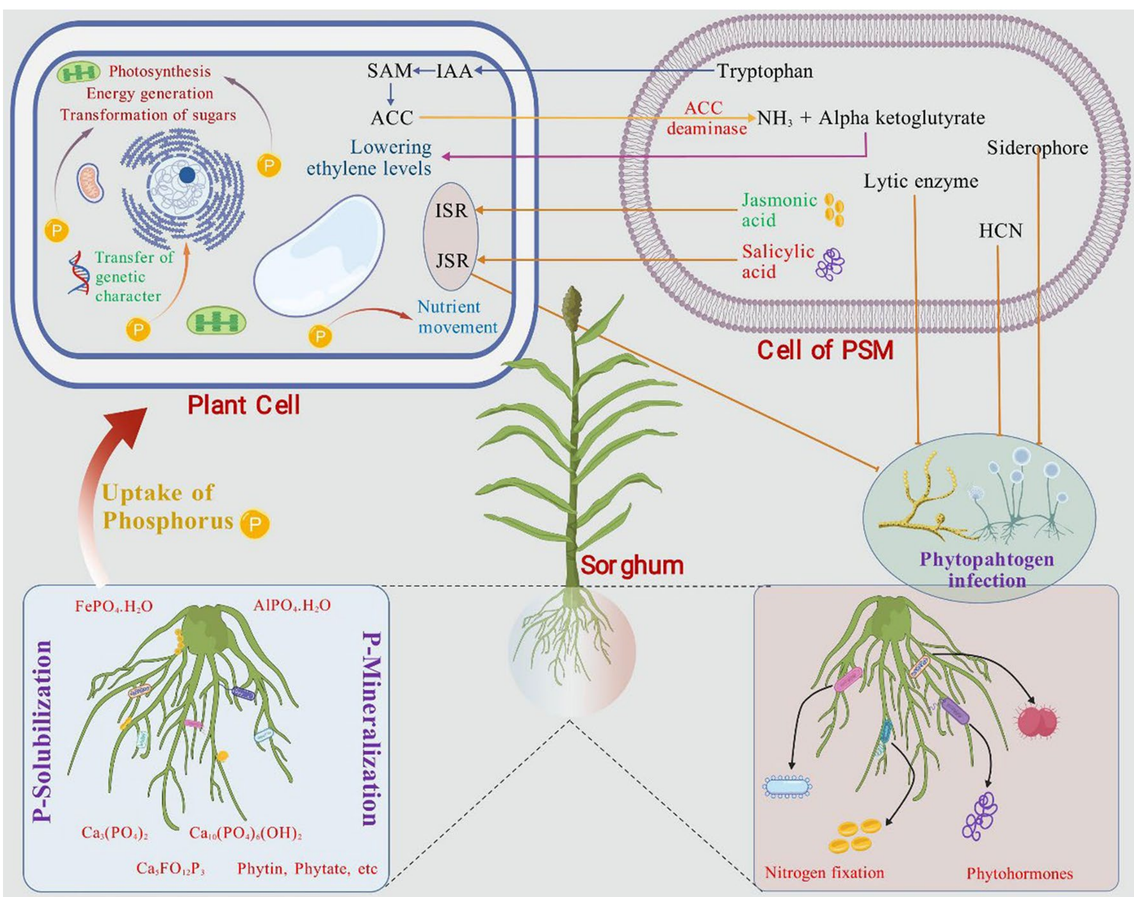


Fig. 3 Mechanisms of phosphorus solubilization and plant growth promotion by phosphate-solubilizing microorganisms (PSM) in sorghum. The illustration depicts the interaction between PSM and sorghum roots, facilitating phosphorus (P) uptake through solubilization and mineralization processes. PSM enhance plant growth by producing phytohormones (indole-3-acetic acid, jasmonic acid, and salicylic acid), lowering ethylene levels via ACC deaminase, and aiding nutrient movement. Additionally, PSM produce siderophores, hydrogen cyanide (HCN), and lytic enzymes that suppress phytopathogen infections. The overall effect is improved nutrient availability, nitrogen fixation, and plant resistance to biotic and abiotic stresses

improve nutrient uptake, root development, and stress tolerance, ultimately enhancing sorghum productivity and sustainability in nutrient-deficient soils.

Mycorrhizal Fungi and Nitrogen-Fixing Bacteria

Mycorrhizal fungi, especially arbuscular mycorrhizal fungi (AMF), form symbiotic relationships with plant roots, creating a mycelial network that extends well beyond the rhizosphere. This network greatly increases the root surface area for absorbing nutrients and water. In sorghum, colonization by AMF has been found to improve phosphorus uptake, which is essential for energy transfer and root development (Mikiciuk et al. 2024). Moreover, AMF-mediated improvements in drought resistance are linked to better osmotic regulation, which helps maintain cell turgor and reduces water loss through transpiration. Nitrogen-fixing bacteria, although traditionally associated with legumes, also play a vital role in cereals like sorghum. Species such as *Rhizobium* and *Bradyrhizobium* form loose associations with sorghum roots, converting atmospheric nitrogen into ammonium, a form readily usable by plants (Markos and Yoseph 2024). This biological nitrogen fixation reduces the need for synthetic nitrogen fertilizers, which are costly and environmentally detrimental. When combined with mycorrhizal fungi, nitrogen-fixing bacteria create a beneficial consortium that further enhances sorghum's nitrogen and phosphorus uptake, improving overall plant health and productivity.

Role of Endophytes in Stress Tolerance

Endophytes are microorganisms, including bacteria and fungi, that live within plant tissues without causing harm. These hidden allies are gaining recognition for their role in enhancing plant resilience to environmental stresses. In sorghum, bacterial endophytes such as *Enterobacter cloacae* have been found to produce 1-aminocyclopropane-1-carboxylate (ACC) deaminase, an enzyme that lowers ethylene levels in stressed plants (Kruasuwan and Thamchaipenet 2018). Ethylene, a stress hormone, inhibits root elongation under drought conditions. By reducing ethylene accumulation, endophytes help maintain root growth and improve water uptake during prolonged dry periods. Fungal endophytes also contribute significantly to sorghum's defense mechanisms. For example, *Fusarium solani*, when present as an endophyte rather than a pathogen, induces the production of antioxidant enzymes like superoxide dismutase and catalase (Li et al. 2022). These enzymes mitigate oxidative stress caused by reactive oxygen species (ROS), which accumulate during heat and drought stress. By enhancing the plant's antioxidant defense system, endophytes reduce cellular damage and improve plant survival under extreme conditions. Furthermore, metagenomic analyses reveal that endophyte populations can vary depending on the sorghum genotype and environmental conditions, indicating that tailored microbial inoculants may be developed for specific climates and cultivars (Benitez-Alfonso et al. 2023). In brief, the roles of PGPMs, mycorrhizal fungi, nitrogen-fixing bacteria, and endophytes in sorghum resilience highlight the importance of harnessing microbial interactions for sustainable agriculture. Future research integrating these microbial partnerships with advanced biotechnological tools offers significant potential to enhance sorghum productivity and adaptability to climate-induced stresses. Combining microbial technologies with precision agriculture can revolutionize food production systems while reducing chemical input dependency and environmental impact.

Plant Biotechnology Tools for Climate Resilience

CRISPR/Cas9 is a gene-editing technology that makes precise genetic changes, creating permanent modifications without adding foreign DNA. In contrast, transgenics involves inserting foreign genes to introduce new traits, like improved stress tolerance and productivity. RNA Interference (RNAi) regulates gene expression by silencing specific genes without integrating foreign DNA. Marker-assisted selection (MAS) is a non-GMO strategy that uses genetic markers to select desirable traits, improving breeding efficiency without direct genetic modification. The Venn diagram (Fig. 4) highlights the relationships between these approaches, distinguishing gene expression modulation, gene silencing, and permanent genetic modifications while differentiating GMO and non-GMO strategies for sorghum improvement. A brief overview of biotechnological tools and their applications in sorghum is presented, highlighting each tool's specific use and underlying benefit explain in Table 3.

Genomic and Transcriptomic Approaches

Genomic and transcriptomic technologies have become indispensable tools in the effort to enhance climate resilience in sorghum. The sequencing of the sorghum genome has provided an essential resource for identifying genes that play a role in environmental stress responses, opening new avenues for improving climate resilience through precise genetic modifications. The *Sorghum bicolour* genome, which was fully sequenced in 2009, contains approximately 34,000 protein-coding genes, Cytochrome P450 domain-containing genes, some are involved in scavenging toxins like those accumulated in response to stress, are abundant in sorghum with 326 versus 228 in rice (Paterson et al. 2009). Recent advances in comparative genomics have enabled the identification of orthologous genes involved in stress tolerance

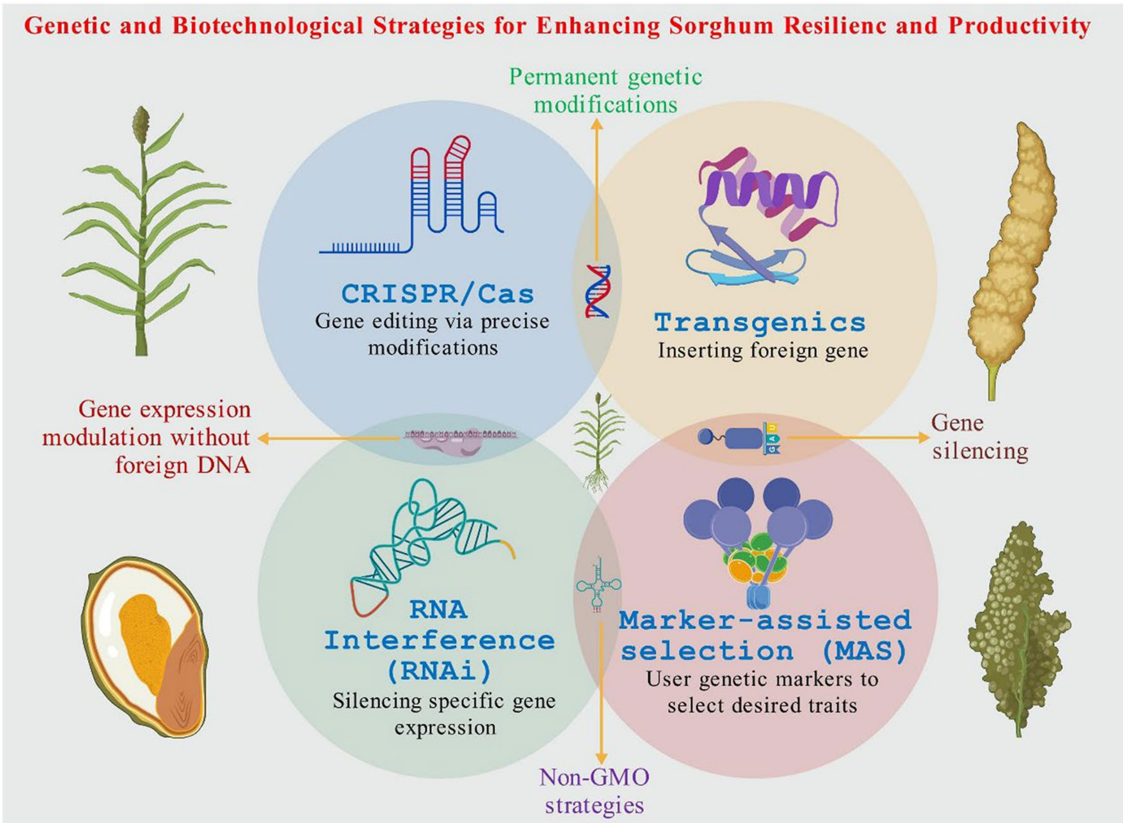


Fig. 4 The figure presents genetic and biotechnological strategies aimed at enhancing sorghum resilience and productivity. CRISPR/Cas (top-left, blue section), A gene-editing technology enabling precise genetic modifications without introducing foreign DNA, leading to permanent genetic modifications; Transgenics (top-right, yellow section), involves the insertion of foreign genes to introduce novel traits, such as stress tolerance and productivity enhancement; RNA Interference (RNAi) (bottom-left, orange section), A technique for gene expression modulation through specific gene silencing, without incorporating foreign DNA; Marker-Assisted Selection (MAS) (bottom-right, purple section), A non-GMO strategy that utilizes genetic markers to select for desirable traits, improving breeding efficiency without direct genetic modification. This Venn diagram structure highlights the relationships between these approaches, distinguishing between gene expression modulation, gene silencing, and permanent genetic modifications, while also differentiating GMO and non-GMO strategies for sorghum improvement

across other cereal crops, offering insights into conserved stress-resilient pathways tabulated in Table 4 (Kamali and Singh 2023).

Transcriptomic analyses have revealed the dynamic gene expression patterns that occur in response to environmental stressors like drought, high salinity, and extreme temperatures. RNA sequencing (RNA-Seq) studies show upregulation of osmotic stress tolerance genes, including dehydration-responsive proteins such as *SbDREB2* (dehydration-responsive element-binding protein 2), and transcription factors like *SbWRKY* and *SbNF-Y* essential for cell integrity (Seok et al. 2023).

Further, multi-omics approaches, integrating genomics, transcriptomics, proteomics, and metabolomics, have led to a more comprehensive understanding of how sorghum modulates its stress tolerance mechanisms at various biological levels (Mwamahonje et al. 2021). The combination of genomic and transcriptomic data has enabled researchers to pinpoint candidate genes for targeted selection, thus facilitating the breeding of sorghum varieties with enhanced resilience to changing climatic conditions.

CRISPR/Cas9 and Genome Editing Applications

CRISPR/Cas9 technology has significantly impacted plant genetic engineering by offering precise tools for genome editing. CRISPR allows for the targeted modification of genes responsible for various traits, including drought tolerance, pest resistance, and improved nutrient use efficiency. In sorghum, CRISPR/Cas9 has been applied to target genes associated with abiotic stress tolerance and metabolic pathways that influence water retention, osmotic adjustment, and photosynthesis (Sami et al. 2021). For instance, CRISPR-mediated knockout of the *DREB2* gene, which encodes a key transcription factor, has resulted in enhanced drought tolerance

Table 3 Biotechnological Tools and Their Applications in Sorghum

Biotechnological Tool	Application	Mechanism/Benefits	References
Genome Editing (CRISPR/Cas9)	Targeted gene editing for stress tolerance, pest resistance, and grain quality	Allows precise modification of specific genes to enhance drought tolerance, heat resistance, and nutritional content	Kumar et al. 2023
Marker-Assisted Selection (MAS)	Selection of superior traits in breeding programs	Accelerates the breeding process by identifying genetic markers linked to desirable traits such as pest resistance and high yield	Somegowda et al. 2024
RNA Interference (RNAi)	Gene silencing for enhancing abiotic stress tolerance	Downregulates the expression of genes associated with stress response mechanisms, improving drought and salt tolerance	Li et al. 2023
Transgenics (Genetic Engineering)	Development of genetically modified sorghum varieties with improved traits	Insertion of foreign genes, such as those encoding for pest resistance or improved nitrogen fixation, enhancing overall plant resilience	Baloch et al. 2023; Nigam et al. 2025
Omics Technologies (Transcriptomics, Proteomics, Metabolomics)	Comprehensive profiling of sorghum's response to environmental stress	Provides insights into gene expression, protein function, and metabolic pathways that enable the identification of stress-responsive genes for breeding	Tu et al. 2023; Mmbando 2024
Synthetic Biology	Engineering microorganisms for enhanced stress tolerance and nutrient uptake	Designing and modifying microbial communities or genes to improve plant growth and nutrient availability, enhancing sorghum's resilience	Kumar and Sindhu 2024
Genomic Selection (GS)	Predicting plant breeding values for complex traits	Uses genomic data to predict the performance of breeding lines for traits like yield, drought tolerance, and disease resistance	Krishnappa et al. 2021; Maulana et al. 2023
Bioinformatics Tools	Analysis of sorghum genome sequences for functional genomics	Facilitates the identification of candidate genes involved in stress tolerance, yield enhancement, and pest resistance	Somegowda et al. 2024

Table 4 Key Genes Regulating Abiotic Stress Responses in Sorghum

Gene(s)	Function	Stress Type(s)	Citation
SbSNAC1	NAC transcription factor conferring drought tolerance	Drought	Abdel-Ghany et al. 2020
SbNAC014, *SbNAC035*, *SbNAC041*	Post-flowering drought tolerance	Drought	Abdel-Ghany et al. 2020
SbDof12, *SbDof19*, *SbDof24*	Early drought response regulators	Drought	Abdel-Ghany et al. 2020
SbWRKY30	Taproot/leaf expression under stress	Drought	Abdel-Ghany et al. 2020
SbWRKY74, *SbWRKY75*, *SbWRKY19*, *SbWRKY5*, *SbWRKY45*, etc	Multi-stage drought stress regulation	Drought	Abdel-Ghany et al. 2020
SbP5CS1, *SbP5CS2*	Proline biosynthesis under drought	Drought	Abdel-Ghany et al. 2020
SbER2-1	Involved in phenylpropanoid metabolism and drought response	Drought	Abdel-Ghany et al. 2020
SbHSF1	Heat shock transcription factor (heat stress)	Heat	Johnson et al. 2014
SbHSF5, *SbHSF13*	Heat-induced under drought conditions	Heat	Johnson et al. 2014
SbHKT1;4	Ion homeostasis under salt and drought stress	Drought, Salinity	Wang et al. 2014
SbHAK2, *SbHAK3*, *SbHAK20*, *SbHAK21*	Potassium transport under abiotic stress	Drought, Heat, Salt, Cold	Anil Kumar et al. 2022
SbWRKY24	Multi-stress responsive WRKY TF	Drought, Salt, Heat	Baillo et al. 2020
DREB1a, *DREB1b* (AP2/ERF family)	Regulate drought-responsive gene expression	Drought	Fracasso et al. 2016
SOD1, *VTC1*, *MDAR1*, *MSRB2*, *ABC1K1*	ROS detoxification and oxidative stress defense	Drought	Abdel-Ghany et al. 2020

by upregulating other stress-related genes in sorghum (Angon et al. 2023). Similarly, editing of the *SbBADH2* gene, involved in proline biosynthesis, has been shown to enhance proline accumulation under osmotic stress, contributing to improved drought resilience (Chen et al. 2024). Another promising application of CRISPR/Cas technology in sorghum is in the modification of the plant's root architecture. By editing genes related to root growth and morphology, such as *PIN* (auxin efflux carrier) and *LBD* (Lateral Organ Boundaries domain), researchers have been able to improve root penetration and water uptake efficiency under water-limited conditions (Li et al. 2021). This is a significant advancement, as root traits play a crucial role in the plant's ability to withstand drought stress. CRISPR-based genome editing also offers the possibility of simultaneously targeting multiple genes associated with complex traits. Recent developments in multiplex CRISPR systems enable the simultaneous editing of multiple stress-related genes, leading to faster improvements in stress resilience (Shelake et al. 2022). Additionally, base-editing and prime-editing technologies, which allow for more precise genetic modifications with fewer off-target effects, are being explored for enhancing traits related to climate resilience in sorghum and other crops (Chen et al. 2024).

CRISPR/Cas9 is a powerful genome-editing tool that allows scientists to make targeted modifications to specific genes in the plant genome (Fig. 5). In the context of drought tolerance, CRISPR/Cas9 can be used to either knock out (disable) genes that negatively affect drought resilience or knock in (introduce) beneficial genetic variations that enhance the plant's ability to withstand water scarcity. Below are some key mechanisms by which CRISPR/Cas9-edited genes confer drought tolerance (Fig. 6).

Targeting Drought-Responsive Genes

Dehydration-responsive element-binding proteins (DREBs) transcription factors activate genes involved in drought stress responses. CRISPR/Cas9 can be used to edit or enhance the expression of *DREB* genes, such as *DREB2A* or *DREB2B*, to upregulate pathways that protect the plant during water deficit. For example, editing these genes can lead to increased production of protective proteins and osmoprotectants, which help maintain cellular integrity under drought conditions. Late embryogenesis abundant (LEA) proteins protect cellular structures and enzymes during dehydration. CRISPR/Cas9 can be used to increase the expression of *LEA* genes, enhancing the plant's ability to retain water and survive prolonged drought.

Modifying Stomatal Regulation

Stomata are tiny pores on plant leaves that regulate water loss through transpiration. CRISPR/Cas9 can target genes involved in stomatal closure, such as those encoding abscisic acid (ABA) receptors (e.g., *PYR/PYL/RCAR* family). By enhancing ABA signaling, plants can close their stomata more efficiently during drought, reducing water loss while maintaining photosynthesis.

Enhancing Osmoprotectant Biosynthesis

Osmoprotectants, such as proline and glycine betaine, help plants maintain cellular osmotic balance under

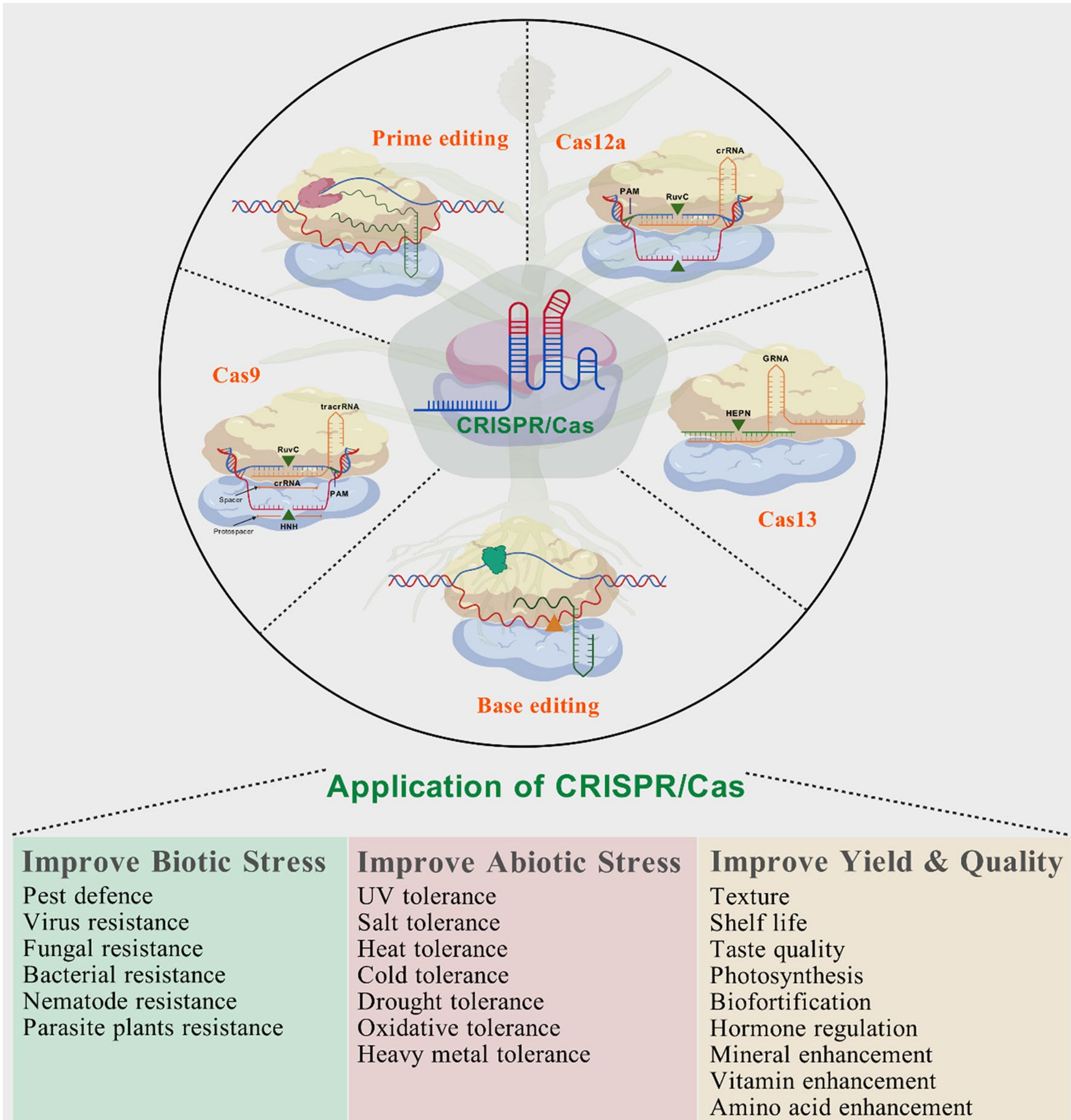


Fig. 5 The image illustrates various CRISPR/Cas genome editing systems as Cas9, Cas12a, Cas13, base editing, and prime editing is used to precisely modify plant genomes. These tools enable a wide range of applications including enhanced resistance to biotic stresses (such as pests, viruses, fungi, and parasites), improved tolerance to abiotic stresses (like drought, salinity, heat, cold, and heavy metals), and the enhancement of yield and quality traits (such as shelf life, nutritional value, and photosynthesis), thereby contributing to sustainable crop improvement

drought stress. CRISPR/Cas9 can edit genes involved in osmoprotectant biosynthesis pathways, such as *P5CS* (pyrroline-5-carboxylate synthase) for proline production. Increased osmoprotectant levels help stabilize proteins, membranes, and cellular structures, improving drought resilience.

Improving Root Architecture

Root systems play a critical role in water uptake during drought. CRISPR/Cas9 can target genes that regulate root growth and architecture, such as *PIN* (auxin efflux carriers) and *LBD* (Lateral Organ Boundaries domain) genes. Editing these genes can promote deeper and more extensive root systems, enabling the plant to access water from deeper soil layers.

Reducing Ethylene Sensitivity

Ethylene is a plant hormone that can inhibit root growth under stress conditions. CRISPR/Cas9 can be used to knock out genes like *EIN3* (ethylene-insensitive 3) or edit

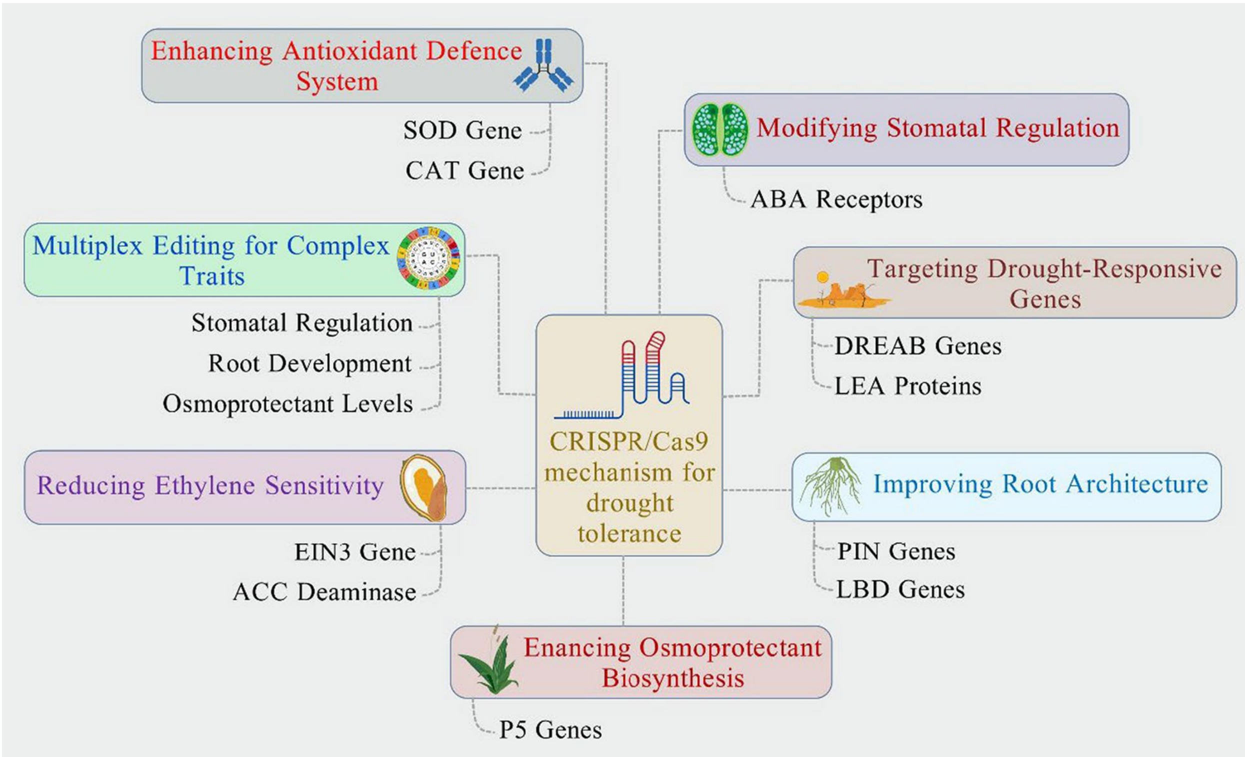


Fig. 6 The schematic diagram highlights CRISPR/Cas9-mediated mechanisms for drought tolerance in plants, including enhancing antioxidant defence (SOD, CAT genes), modifying stomatal regulation (ABA receptors), targeting drought-responsive genes (DREB, LEA proteins), improving root architecture (PIN, LBD genes), enhancing osmoprotectant biosynthesis (P5 genes), reducing ethylene sensitivity (EIN3, ACC deaminase), and multiplex editing for complex traits (stomatal regulation, root development, osmoprotectant levels). These targeted edits contribute to improved stress resilience and drought adaptation

genes encoding ACC deaminase, an enzyme that reduces ethylene levels. Lower ethylene sensitivity allows roots to continue growing even under drought stress, improving water and nutrient uptake.

Enhancing Antioxidant Defence Systems

Drought stress often leads to the accumulation of reactive oxygen species (ROS), which can damage cellular components. CRISPR/Cas9 can target genes involved in antioxidant production, such as *SOD* (superoxide dismutase) and *CAT* (catalase), to enhance the plant's ability to scavenge ROS and protect cells from oxidative damage.

Multiplex Editing for Complex Traits

CRISPR/Cas9 allows for the simultaneous editing of multiple genes, enabling the development of plants with enhanced drought tolerance through a combination of traits. For instance, a single sorghum plant can be engineered for better stomatal regulation, deeper roots, and increased osmoprotectant levels, providing a multi-layered defense against drought (Wei et al. 2024).

Marker-Assisted Selection (MAS) in Breeding Programs

Marker-assisted selection (MAS) has emerged as a powerful tool in plant breeding, allowing for the rapid selection of desirable traits without the need for phenotypic screening. In sorghum, MAS has been particularly useful for the selection of traits related to climate resilience, including drought tolerance, heat stress resistance, and early maturity. Key genetic markers have been identified for traits such as stay-green, a characteristic that delays senescence and maintains photosynthetic capacity under drought conditions. Stay-green is associated with several QTLs, including *Stg1* and *Stg2*, and markers linked to these loci have been used in breeding programs to develop drought-resistant sorghum varieties (Abebe et al. 2021). However, markers associated with root architecture traits, such as root length and density, have been integrated into MAS strategies to select for better water uptake and drought tolerance (Tao et al. 2024).

Molecular markers linked to heat tolerance have been harnessed to enhance sorghum's resilience to withstand global temperatures. The identification of heat-tolerant alleles, particularly those associated with the *HSP70* (heat shock protein) gene, has expedited the development of heat-resistant varieties through MAS (Mulaudzi-Masuku et al. 2015). Integrating MAS with advanced breeding tools, such as genomic selection (GS) and high-throughput phenotyping, significantly improves the efficiency and precision of sorghum breeding programmes (Muleta et al. 2019). Combined with high-throughput phenotyping techniques, which use automated systems to measure plant traits under various stress conditions, MAS has become a powerful strategy for accelerating the breeding of climate-resilient sorghum.

Integrating Microbial and Biotechnological Approaches

The integration of microbial inoculation with plant biotechnology has garnered significant attention in recent years as a means of enhancing crop resilience, particularly under the stressors of climate change. Beneficial plant-associated microbes, including bacteria, fungi, and other microorganisms, contribute to plant growth by enhancing nutrient uptake, improving stress tolerance, and reducing susceptibility to pathogens. When combined with cutting-edge biotechnological tools such as genetic modification and genome editing, these interactions offer a synergistic approach to improving plant performance, especially in climate-resilient crops like sorghum.

Synergies between Biotechnology and Microbial Inoculation

Recent discovery in microbial biotechnology is the identification of specific microbial strains that enhance plant–microbe signalling pathways. For example, researchers have identified *Bacillus* and *Pseudomonas* species that produce extracellular polysaccharides (EPS) that not only enhance root colonization but also help plants retain water under drought conditions (Naseem et al. 2024). Genetically modified plants that increase root exudate secretion can enhance the recruitment of beneficial microbes (Afridi et al. 2024). This improves microbial inoculant effectiveness in boosting drought resistance, showcasing the importance of biotechnological advancements in this area.

Moreover, plant biotechnology tools like CRISPR/Cas9 have enabled more precise genetic modifications to improve the plant's compatibility with beneficial microbes. For example, recent studies have demonstrated the successful modification of sorghum to upregulate genes involved in the formation of root nodules, allowing it to form more favourable interactions with nitrogen-fixing bacteria such as *Rhizobium* species (Guo et al. 2023). This is particularly important for improving the nitrogen-use efficiency of crops, reducing the need for synthetic fertilizers, and promoting sustainable farming practices.

Sorghum and PGPR for Drought and Salinity Tolerance

The integration of biotechnology and microbial inoculants has shown remarkable potential across various cereal crops, with sorghum being one of the primary beneficiaries of these innovations. A study demonstrated that sorghum plants inoculated with a consortium of PGPR, including *Azospirillum brasilense*, *Bacillus subtilis*, and *Pseudomonas putida*, exhibited improved drought tolerance. These microorganisms secrete phytohormones, such as IAA, and enhance root development, enabling better water uptake under arid conditions. In combination with a genetically engineered sorghum variety that overexpresses drought-resilient genes like s*DREB1* and *RD29A*, the plants showed superior growth and grain yield under drought conditions (Carlson et al. 2020; Grover et al. 2021). This case exemplifies how combining microbial inoculation with genetic enhancements can provide a holistic solution for sustainable sorghum production in water-scarce regions.

Recent Discoveries and Future Directions

In the past few years, several exciting breakthroughs have emerged in the field of plant–microbe interactions and biotechnology. One such discovery is the identification of novel microbial consortia that can be tailored to specific environmental conditions. For example, researchers have found that a combination of *Bacillus*, *Pseudomonas*, and *Rhizobium* species can be optimized to promote both drought and nutrient stress tolerance simultaneously (Chieb and Gachomo 2023). These microbial communities are being incorporated into biotechnological formulations that offer practical, scalable solutions for farmers in water-scarce regions. Another recent development is the use of synthetic biology to create microbial biofertilizers that can be deployed alongside genetically modified crops for maximum efficiency. By designing microbes to produce beneficial metabolites, such as phytohormones or antioxidants, researchers are creating designer microbial strains that can boost crop resilience to environmental stressors (Ali et al. 2024; Nadarajah and Abdul Rahman 2023). These microbial formulations, combined with genetically modified crops, promise to enhance productivity and sustainability in a variety of crop systems.

Challenges and Future Perspectives

Integrating microbial inoculation and biotechnological advancements still faces challenges, including the long-term stability of microbial inoculants in varying environments. Variability in microbial communities and potential interaction with native microbiota hinder large scale applications. Furthermore, the complex regulatory landscape for genetically modified crops and microbial products requires careful monitoring and approval. Future research should focus on optimizing microbial formulations for specific regions and crops, ensuring that they complement the genetic modifications of plants. Furthermore, improving the scalability and cost-effectiveness of these integrated systems will be crucial for widespread adoption in commercial agriculture.

Future Perspectives and Challenges in Advancing Climate-Resilient Sorghum

Emerging Technologies in Microbial Research

The role of beneficial microorganisms in enhancing the resilience of crops, including sorghum, under climate-induced stresses has garnered considerable interest. With advancements in microbial research, new technologies are emerging that offer opportunities to optimize microbial interactions for enhanced crop productivity. One of the most promising developments is the application of metagenomics and microbiome-based approaches, which enable the comprehensive identification and functional characterization of microbial communities associated with sorghum plants (Nwachukwu et al. 2022). These technologies provide insights into the genetic diversity of beneficial microbes, such as plant growth-promoting rhizobacteria (PGPR), endophytes, and mycorrhizal fungi, and their contributions to plant growth and stress tolerance. Furthermore, advances in synthetic biology have opened new avenues for designing tailored microbial consortia. By engineering microbial strains with specific functional traits, such as drought tolerance, nitrogen fixation, or pathogen resistance, it is possible to create optimized microbial solutions that work synergistically with sorghum plants to improve yield and resilience (Mikiciuk et al. 2024). This targeted manipulation of microbial communities through bioinformatics tools and high-throughput screening techniques allows for the development of next-generation biofertilizers and biostimulants that can be deployed on a large scale. Genome editing technologies, particularly CRISPR/Cas9, hold great potential for manipulating microbial genomes to enhance their beneficial effects on plants. For example, editing microbial genes to improve their nitrogen-fixing capacity or stress tolerance could result in more efficient interactions with sorghum, thus reducing the need for chemical fertilizers and enhancing sustainability (Li et al. 2024). While these technologies are still in their early stages, their application in crop protection and resilience is promising.

Policy and Funding Considerations for Climate-Resilient Crops

To facilitate the widespread adoption of emerging technologies for climate-resilient crops, particularly sorghum, supportive policies and funding mechanisms are essential. Climate-resilient crops are integral to global food security, particularly in regions prone to drought and other environmental stresses. However, the development and deployment of innovative crop varieties, including those enhanced by microbial interactions and biotechnology, require substantial financial investment in research and development (R&D), infrastructure, and education. Governments and international organizations need to prioritize funding for climate-smart agriculture through public–private partnerships, grants, and collaborations between research institutions and industry stakeholders. Funding should not only support basic research into plant–microbe interactions but also the scaling of solutions that can be integrated into real-world agricultural systems. Policy frameworks that encourage the use of biotechnological advancements, such as regulatory processes for genetically modified organisms (GMOs), need to be streamlined to ensure the timely release and adoption of climate-resilient crops. Regulatory policies must strike a balance between fostering innovation and ensuring safety and sustainability. Clear guidelines for field trials, environmental impact assessments, and consumer acceptance are vital to the successful commercialization of climate-resilient sorghum varieties. Public awareness campaigns and education on the benefits of biotechnology, including microbial solutions and genetic engineering, can also help mitigate public concerns and facilitate the acceptance of these technologies.

Challenges in Scaling Solutions for Farmers

Despite the promising potential of plant biotechnology and microbial interactions to enhance sorghum's resilience, several challenges remain in scaling these solutions to small holder and resource-poor farmers who are most vulnerable to climate change. One of the primary barriers is the access to and affordability of advanced technologies, including genetically improved sorghum varieties and microbial inoculants. The cost of developing and distributing these innovations can be prohibitive, especially in low-income countries where farmers may lack the financial resources to purchase these inputs (Ruzzante et al. 2021). In addition, the effective application of microbial solutions requires a thorough understanding of the environmental conditions and the specific microbial strains that are best suited for particular agro-climatic zones. Farmers may not have the necessary knowledge or access to extension services that could help them implement these solutions effectively. There is also the challenge of ensuring the consistency and reliability of microbial inoculants under field conditions, as microbial communities can be highly variable depending on soil composition, climate, and other factors (O'Callaghan et al. 2022). Another significant challenge is the scalability of biotechnological solutions. While laboratory and greenhouse trials may show promising results, translating these innovations to large-scale agricultural systems presents logistical and economic hurdles. The production, distribution, and storage of microbial inoculants, for example, require careful consideration to ensure that they remain viable and effective over extended periods and under varying environmental conditions. Finally, the adoption of climate-resilient crops and microbial technologies will require strong collaboration between government agencies, research institutions, the private sector, and local farming communities. Training programs, extension services, and farmer-to-farmer networks can facilitate the dissemination of knowledge and ensure that the benefits of these innovations reach the end-users. Furthermore, policies that support farmer access to technology, such as subsidies, tax incentives, and financing options, will be essential for overcoming these barriers.

Future Recommendations

To strengthen the development of climate-resilient sorghum, future research should focus on integrating advanced genomic and biotechnological tools such as CRISPR/Cas-based genome editing, transcriptome profiling, and multi-omics approaches to unravel complex stress-responsive pathways. Emphasis should also be placed on decoding the functional role of beneficial plant-associated microbes, particularly plant growth-promoting rhizobacteria (PGPR) and endophytes, in enhancing sorghum's tolerance to abiotic stresses like drought, salinity, and heat. Isolation and characterization of novel microbial consortia, combined with host genotype screening, may lead to microbe-assisted breeding strategies. Long-term field trials under variable climate conditions, supported by systems biology and precision agriculture technologies, will be essential for validating lab-scale outcomes. Establishing a shared genomic–microbiome database for sorghum could further accelerate targeted interventions and cultivar development tailored to specific agro-climatic zones.

Conclusion

The integration of plant biotechnology and microbial interactions offers a cutting-edge approach to enhancing the climate resilience of sorghum, a crucial cereal crop for global food security. The current body of research underscores the pivotal role of beneficial microbes such as PGPR, mycorrhizal fungi, and endophytic bacteria, in improving sorghum's tolerance to both abiotic and biotic stress. These microbes contribute to enhanced nutrient acquisition, particularly nitrogen and phosphorus, water-use efficiency, and overall plant vigor, which are critical for maintaining crop productivity under fluctuating environmental conditions. Concurrently, biotechnological advancements have significantly improved the molecular tools available to enhance sorghum's intrinsic resilience. Genetic engineering approaches, including CRISPR/Cas9 and transgenic breeding, have led to sorghum varieties with traits for abiotic stress tolerance, improved grain quality, and higher yields. These modifications enhance plant performance alongside beneficial microbial communities, while synthetic biology aids in designing microbial consortia and crops for diverse environments.

The interaction between sorghum and its associated microbes is not merely a passive relationship but a dynamic process regulated by a network of chemical signals and physiological responses. These interactions involve intricate feedback loops, where plants influence the composition and activity of root-associated microbes, and vice versa. Such mutualistic relationships are enhanced by the application of biotechnology, which can accelerate the establishment of beneficial microbial communities in the rhizosphere and endosphere of sorghum. PGPR strains, such as *Azospirillum*, *Bacillus*, and *Pseudomonas*, have been shown to produce phytohormones, such as IAA and cytokinins, which not only promote plant growth but also induce systemic resistance to pathogens. Furthermore, AMF enhances nutrient uptake, especially under nutrient-poor and drought-prone conditions, facilitating better root colonization and soil–water retention. In supposition, the combination of plant biotechnology and microbial interactions represents a powerful strategy for advancing climate-resilient sorghum. This integrated approach, underpinned by continuous research and technological advancements, holds the potential to meet the challenges posed by climate change while ensuring the sustainability of global food systems. By aligning microbial and biotechnological interventions, future sorghum cultivation can be made more productive, environmentally sustainable, and resilient to the impacts of climate variability.

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Author contributions
A.K.S. and A.R. wrote the main manuscript text, J.J, X.L, M.U. and P.M. prepared figures and tables. A.Z., J.Z., R.P.S., L.L., S.C., S.Y., Y.Z., and X.X. help in editing and reviewed the manuscript.

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Data availability
No datasets were generated or analysed during the current study.

Declarations

Competing interests
The authors declare no competing interests.

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